

Theorem (Cauchy–Schwarz). For any real numbers $a_1, \dots, a_n$ and $b_1, \dots, b_n$,

$$\left(\sum_{i=1}^n a_i b_i\right)^2 \leq \left(\sum_{i=1}^n a_i^2\right) \left(\sum_{i=1}^n b_i^2\right).$$

Equality holds iff there exists $\lambda \in \mathbb{R}$ such that $a_i = \lambda b_i$ for every i (including the trivial case $a_i = b_i = 0$ for all i).

Proof. 1. **Quadratic function.** Define

$$f(t) = \sum_{i=1}^n (a_i - tb_i)^2, \quad t \in \mathbb{R}.$$

Each summand is a square, hence $f(t) \geq 0$ for all real t .

2. **Expansion.** For a single index i ,

$$(a_i - tb_i)^2 = a_i^2 - 2a_i b_i t + b_i^2 t^2.$$

Summing over i gives

$$\begin{aligned} f(t) &= \sum_{i=1}^n a_i^2 - 2t \sum_{i=1}^n a_i b_i + t^2 \sum_{i=1}^n b_i^2 \\ &= \left(\sum_{i=1}^n b_i^2\right) t^2 - 2\left(\sum_{i=1}^n a_i b_i\right) t + \sum_{i=1}^n a_i^2. \end{aligned}$$

Thus $f(t) = At^2 + Bt + C$ with

$$A = \sum_{i=1}^n b_i^2, \quad B = -2 \sum_{i=1}^n a_i b_i, \quad C = \sum_{i=1}^n a_i^2.$$

Note that $A \geq 0$.

3. **Non-negativity of a quadratic.** A real quadratic $At^2 + Bt + C$ with $A > 0$ satisfies $At^2 + Bt + C \geq 0$ for all t iff its discriminant is non-positive:

$$B^2 - 4AC \leq 0.$$

(If $A = 0$ the polynomial is linear; the degenerate cases are handled in step 6.)

4. **Discriminant.** Substituting A, B, C ,

$$\begin{aligned} B^2 - 4AC &= \left(-2 \sum_{i=1}^n a_i b_i\right)^2 - 4\left(\sum_{i=1}^n b_i^2\right)\left(\sum_{i=1}^n a_i^2\right) \\ &= 4\left[\left(\sum_{i=1}^n a_i b_i\right)^2 - \left(\sum_{i=1}^n a_i^2\right)\left(\sum_{i=1}^n b_i^2\right)\right]. \end{aligned}$$

5. **Deriving the inequality.** Since $f(t) \geq 0$ for all t , the discriminant must satisfy $B^2 - 4AC \leq 0$. Dividing the expression in step 4 by 4 yields

$$\left(\sum_{i=1}^n a_i b_i\right)^2 \leq \left(\sum_{i=1}^n a_i^2\right) \left(\sum_{i=1}^n b_i^2\right),$$

which is the Cauchy–Schwarz inequality.

6. **Degenerate cases.**

- If $\sum_{i=1}^n a_i^2 = 0$, then $a_i = 0$ for all i and both sides of the inequality are 0.
- If $\sum_{i=1}^n b_i^2 = 0$, then $b_i = 0$ for all i and again the inequality reduces to $0 \leq 0$.

Thus the inequality holds in all possible configurations.

7. **Equality condition.** Equality occurs precisely when $B^2 - 4AC = 0$, i.e. when the quadratic $f(t)$ has a double root. For $A > 0$ this happens iff there exists $\lambda \in \mathbb{R}$ such that

$$a_i - \lambda b_i = 0 \quad \text{for all } i,$$

or equivalently $a_i = \lambda b_i$ for every i . Indeed, solving $f'(\lambda) = 0$ gives

$$\lambda = \frac{\sum a_i b_i}{\sum b_i^2},$$

and $f(\lambda) = 0$ forces $a_i = \lambda b_i$ for each index. Conversely, if $a_i = \lambda b_i$ for all i , then each summand $(a_i - \lambda b_i)^2$ is zero, so $f(\lambda) = 0$ and the discriminant vanishes, yielding equality in the Cauchy–Schwarz inequality.

8. **Conclusion.** For arbitrary real sequences $\{a_i\}_{i=1}^n$ and $\{b_i\}_{i=1}^n$ the inequality

$$\left(\sum_{i=1}^n a_i b_i\right)^2 \leq \left(\sum_{i=1}^n a_i^2\right) \left(\sum_{i=1}^n b_i^2\right)$$

holds, with equality exactly when the two vectors are linearly dependent (or one of them is the zero vector). $\square$

$\square$